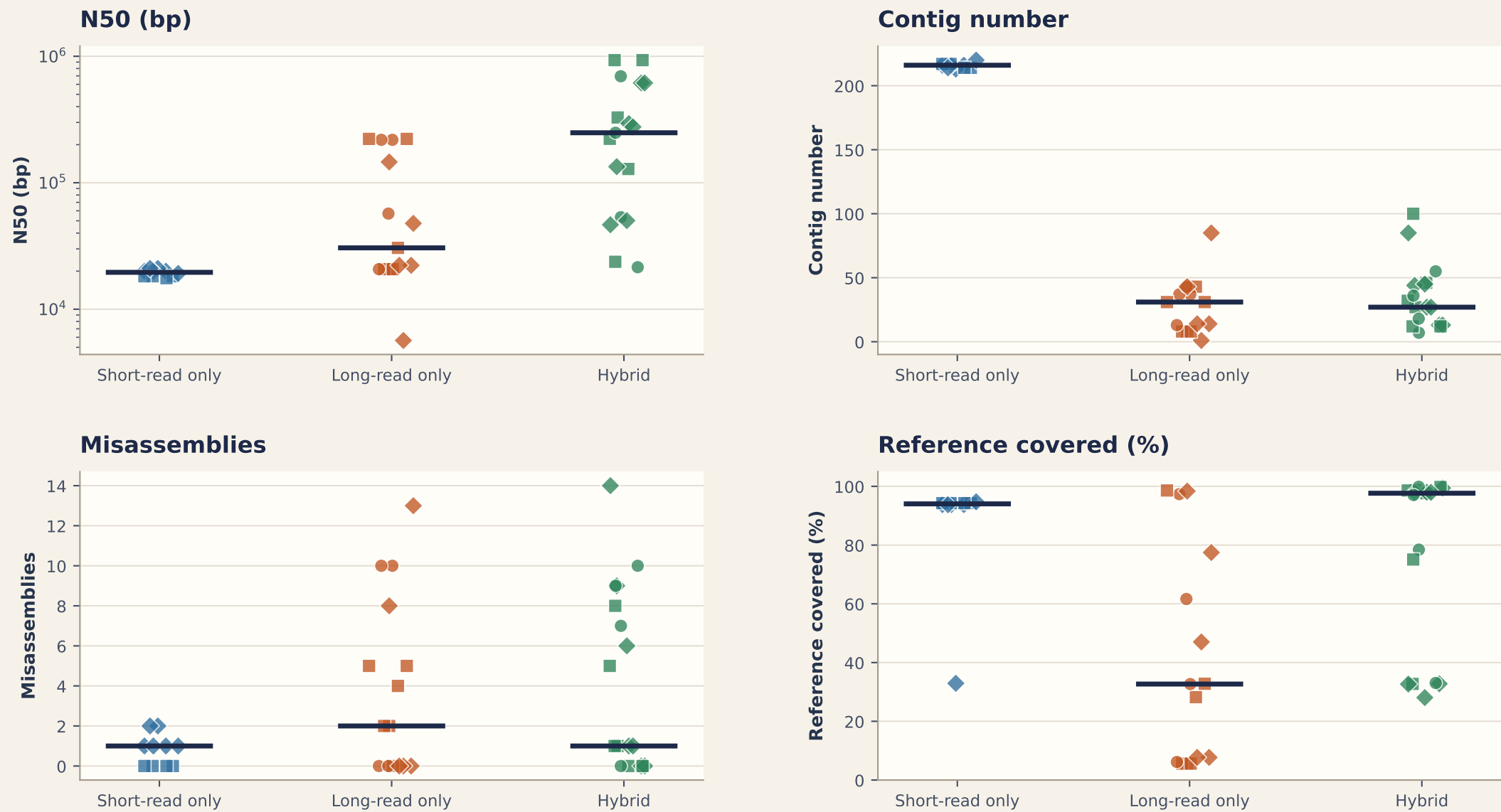


Ireland 2026: Assembly Strategy Trade-offs

Core comparison across 42 complete assembly evaluations. Lines mark the median within each strategy.

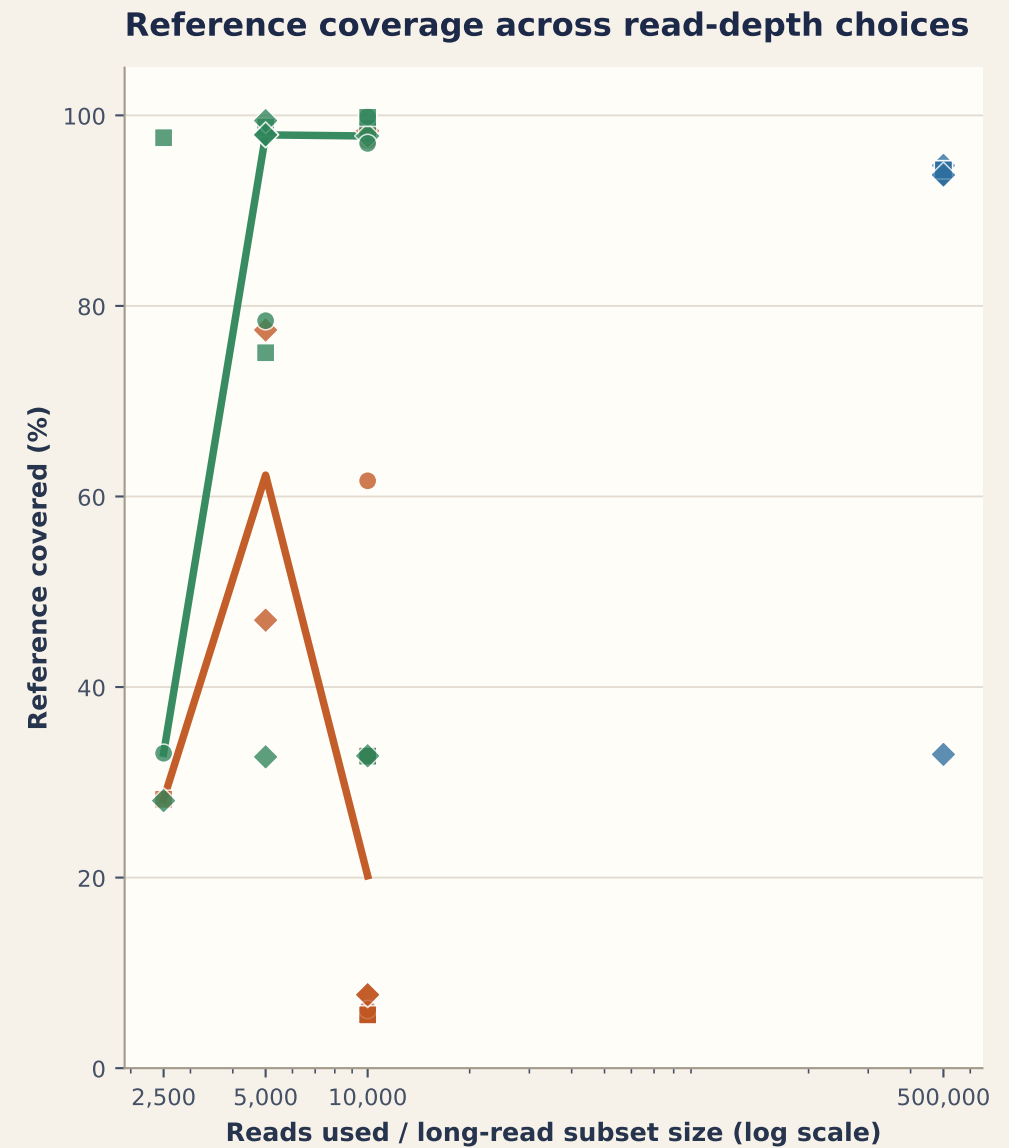
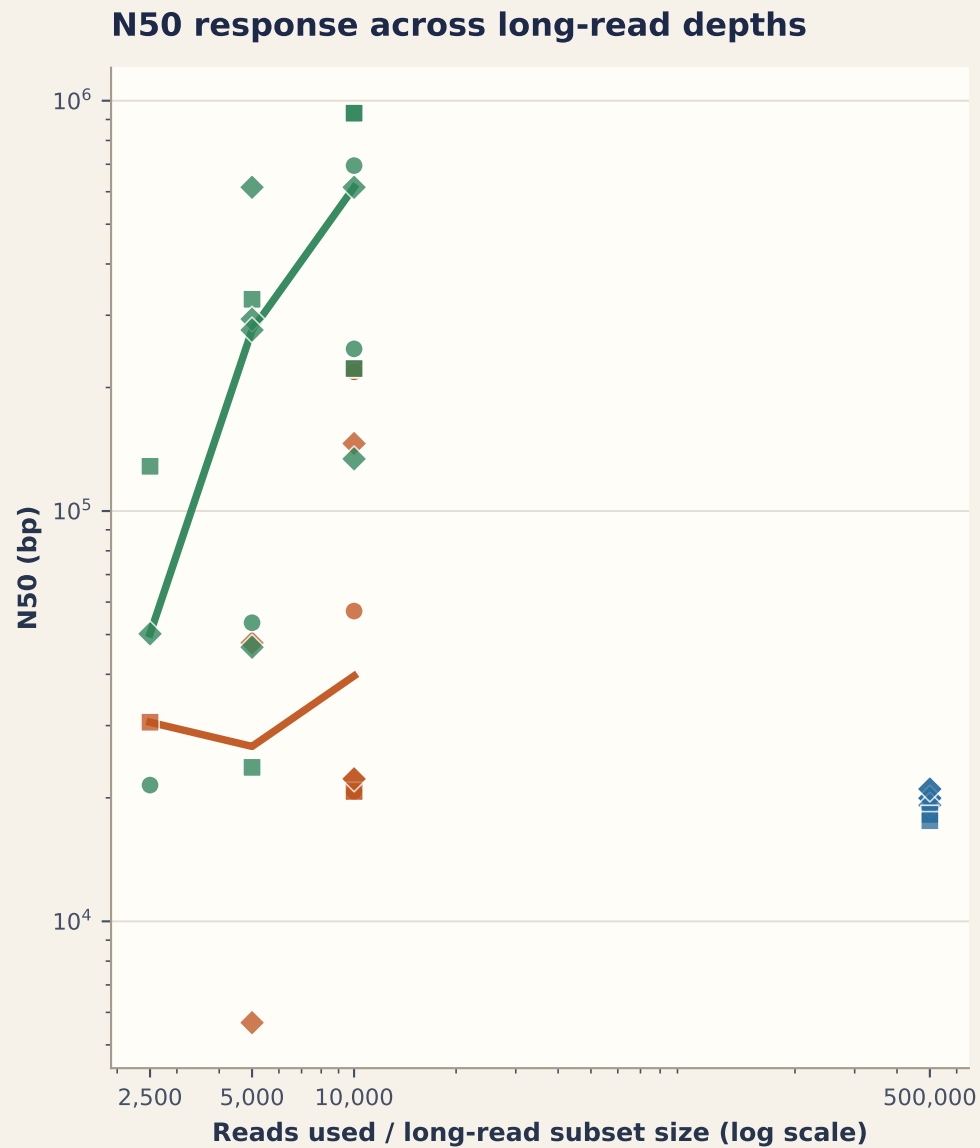


At a glance: hybrid assemblies had the highest median N50 (248,450 bp) and reference coverage (97.7%). Short-read assemblies retained high coverage (94.0%) but were more fragmented (median 216 contigs).

Read-depth response

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Each point is an assembly. Colour identifies strategy, symbol identifies random seed, and lines show median performance at each depth.



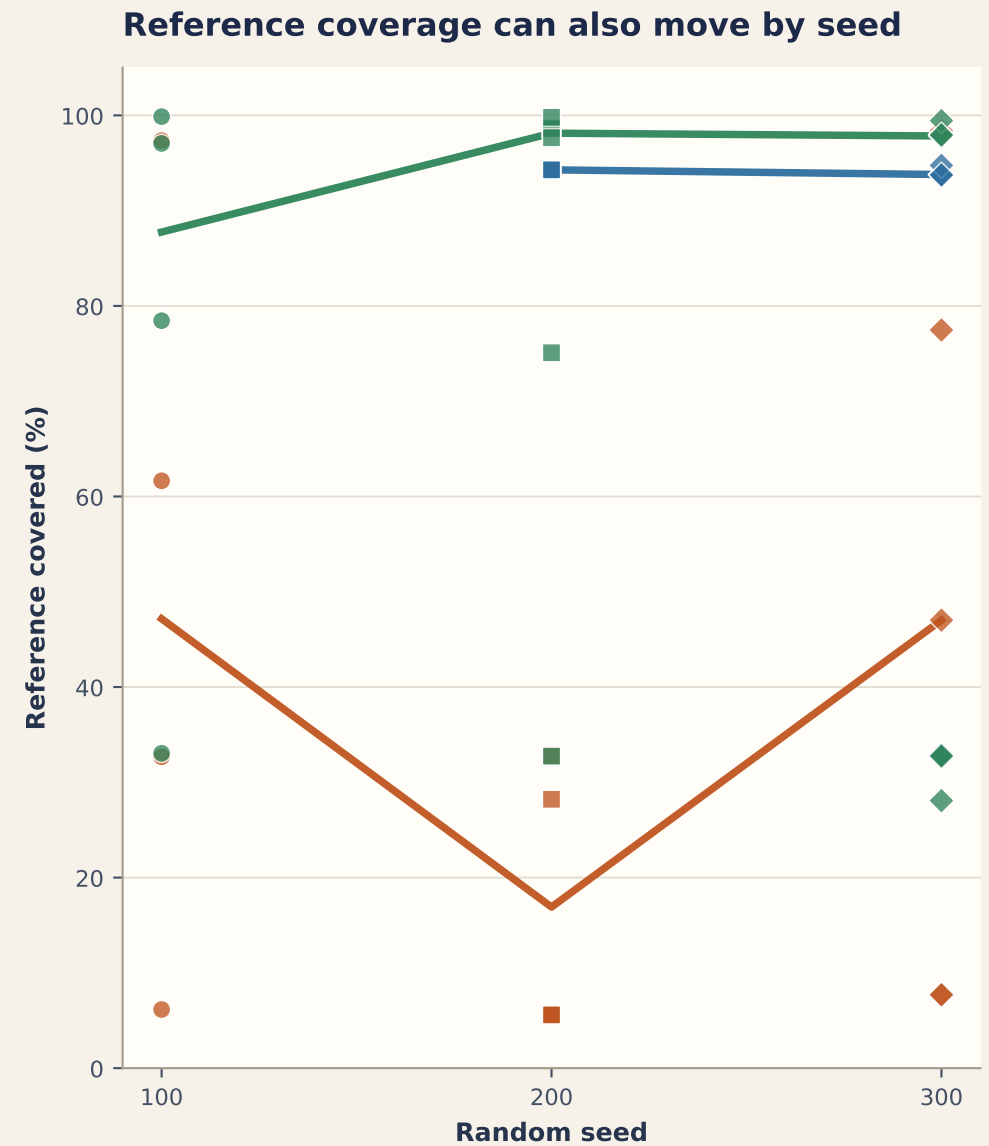
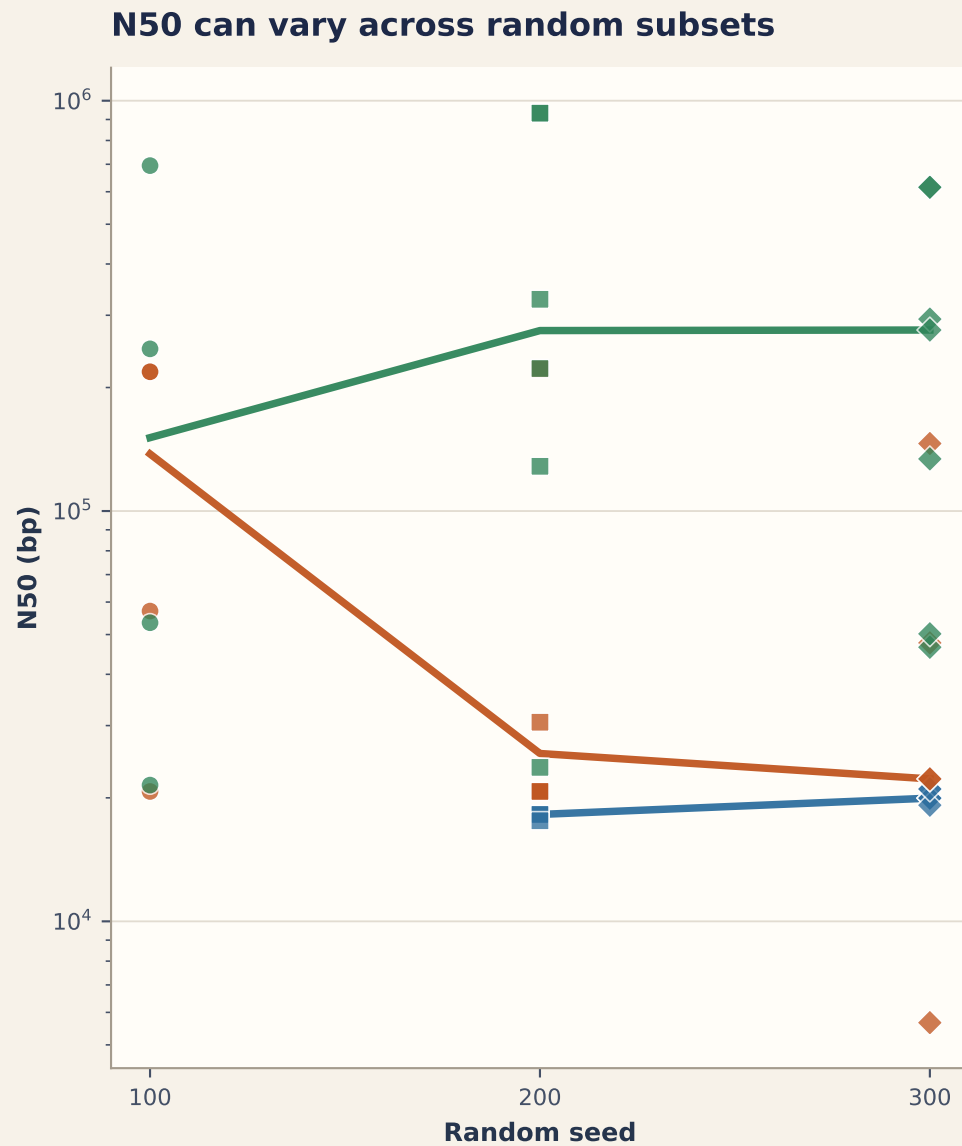
Interpret with care: depth is only one part of the workflow. Assembly strategy, assembler, and the random read subset can all shift performance.

■ Short-read only ■ Long-read only ■ Hybrid ● Seed 100 ■ Seed 200 ◆ Seed 300

Seed sensitivity

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Changing the random subset seed can alter an assembly result. Thick lines show the median within each strategy and seed.



Teaching point: compare strategies using repeated runs where possible; a single run can combine both biological and sampling variation.

■ Short-read only ■ Long-read only ■ Hybrid ● Seed 100 ■ Seed 200 ◆ Seed 300